

STT Assignment

State Transition Technique 1: User Login System States: Logged Out, Logged In, Account Locked

S. No	Current State	Event / Input	Next State	Expected Result
1	Logged Out	Enter valid credentials	Logged In	User accesses dashboard
2	Logged Out	Enter invalid credentials (< 3 attempts)	Logged Out	Error message shown
3	Logged Out	Enter invalid credentials (3rd attempt)	Account Locked	Lockout notification displayed
4	Account Locked	Request password reset	Logged Out	Reset email sent
5	Logged In	Click Logout	Logged Out	Session terminated

State Transition Technique 2: E-Commerce Order Lifecycle States: Created, Paid, Shipped, Delivered, Cancelled

S. No	Current State	Event / Input	Next State	Expected Result
1	Created	Successful payment	Paid	Confirmation email sent
2	Created	Cancel order	Cancelled	Order marked as cancelled
3	Paid	Dispatch items	Shipped	Tracking number generated
4	Paid	Refund requested	Cancelled	Payment refunded
5	Shipped	Package delivered	Delivered	Status updated to delivered

State Transition Technique 3: ATM Cash Withdrawal States: Idle, Card Inserted, PIN Verified, Transaction Complete

S. No	Current State	Event / Input	Next State	Expected Result
1	Idle	Insert debit card	Card Inserted	Prompt for PIN
2	Card Inserted	Enter correct PIN	PIN Verified	Display main menu
3	Card Inserted	Enter incorrect PIN (3 times)	Idle	Card retained/ejected
4	PIN Verified	Select amount & withdraw	Transaction Complete	Cash dispensed & receipt printed
5	Transaction Complete	Eject card	Idle	Screen resets to default

State Transition Technique 4: Online Ticket Reservation States: Available, Reserved (Pending Payment), Booked, Expired

S. No	Current State	Event / Input	Next State	Expected Result
1	Available	Select seat	Reserved	10-minute timer starts
2	Reserved	Complete payment	Booked	Ticket issued
3	Reserved	Timer expires (10 mins)	Expired	Seat released to pool
4	Expired	System refresh	Available	Seat back on market
5	Booked	Cancel booking	Available	Refund processed per policy

State Transition Technique 5: Hotel Room Booking System States: Vacant, Occupied, Cleaning, Under Maintenance

S. No	Current State	Event / Input	Next State	Expected Result
1	Vacant	Guest check-in	Occupied	Keycard issued
2	Occupied	Guest check-out	Cleaning	Housekeeping notified
3	Cleaning	Housekeeping completed	Vacant	Room ready for guest
4	Vacant	Report hardware issue	Under Maintenance	Blocked from booking
5	Under Maintenance	Repair complete	Cleaning	Sent for final inspection